Solution to “Ketch up” on the KenKen

Edward Wu

NOTE: For this solution, we call the columns A through I from left to right and the rows 1 through 9 from top to bottom. Then the squares $B3$ and $B4$ make up the 8+ clue in the second column. A cell is defined to be the intersection of a row and column.

The solution has been broken into sizable parts. Read at your own pace.

When asked to “recall,” the author has supplemented subscripts indicating from which previous part this detail has been proved. $_{\text{This is a subscript.}}$ Superscripts denote footnotes.

Images have been provided at certain points, for the reader’s convenience.

Part I

KenKens are an interesting type of puzzle because you need somewhere to start. This probably is $\frac{1}{4}$ of the puzzle, because you have no clues when you begin, and you have to search for where to start. It’s easy to continue working around the grid once you have your first number. Although the six ? clues give fewer clues for the entire puzzle, sometimes not all the clues are necessary, which is why the puzzle is still completable despite the absence of six clues. In fact, this will help determine where to start, as there are now six less clues to consider.

Not sure if there is a second place to begin, but the place where I began is in Row 2. Notice in Row 2 there are a 3÷ and a 18× clue. Both of them have a limited number of possibilities. The 3÷ clue can be $(1, 3)$, $(2, 6)$, or $(3, 9)$. The 18× clue can be $(2, 9)$ or $(3, 6)$. Notice the contradiction?

If the 18× clue is $(3, 6)$, then the 3÷ clue has no possibilities because neither 3 nor 6 can be in the 3÷ clue. Then the 18× clue must be $(2, 9)$ in some order. Since the 2 and 9 are both used, the 3÷ must be $(1, 3)$ in some order.

This is the first major realization of the puzzle.

Part II

The next point of attention is the 48× clue in Row 1. Because the entire clue is in Row 1, there cannot be any repeats. The only possibilities are $(1, 6, 8)$, $(2, 3, 8)$, $(2, 4, 6)$. You can prove these are all the possibilities because there must be a 6 or an 8. The next biggest factor of 48 is 4, with the largest product you can make with 4 being the largest of three factors is $2 \times 3 \times 4 = 24$.

This may not seem very important, but the $14+$ clue that spans four cells is. The average of these four distinct digits is 3.5, which is pretty concerning since the average is expected to be 5. How many ways can you make 14? $(1, 2, 3, 8)$, $(1, 2, 4, 7)$, $(1, 2, 5, 6)$, $(1, 3, 4, 6)$, $(2, 3, 4, 5)$ are all the possibilities. Notice that only one of these combinations does not have a 1, and only one does not have a 2. Use your skills from Part I to determine which is the only possible outcome from this row.

Did you get it? The single possibility for $14+$ without a 2 is $(1, 3, 4, 6)$. If the $48\times$ clue is $(2, 3, 8)$ or $(2, 4, 6)$ then the 2 will not be in the $14+$. Then the $14+$ is $(1, 3, 4, 6)$. However, no matter which of the two triplets for $48\times$ we use, the 3 or 4 will be used. This is a contradiction. Thus, the $48\times$ is $(1, 6, 8)$ and the $14+$ is $(2, 3, 4, 5)$.

This is the second major realization of the puzzle.

Part III

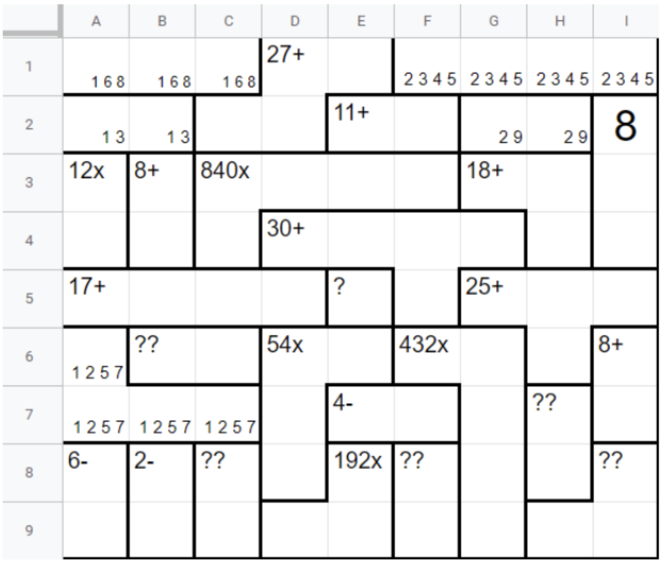
Now consider Rows 1 and 2 together. So far, we have deduced that $48\times$ is $(1, 6, 8)$, the $14+$ is $(2, 3, 4, 5)$, the $3\div$ is $(1, 3)$, and the $18\times$ is $(2, 9)$. Because each digit from 1 to 9 appears once in each row, the sum of the numbers in each row is 45. Two rows add to 90. Thus, in the first two rows, the digits in $48\times$ add to 15. The digits in $3\div$ add to 4. The digits in $18\times$ add to 11. Thus, cell $I2$ must be $90 - 15 - 27 - 14 - 4 - 11 - 11 = 8$.

This is the third major realization of the puzzle.

Part IV

You probably already noticed the $70\times$ that spans Rows 6 and 7 must be made up by $(1, 2, 5, 7)$ because 10 is not an option as the grid is 9×9.

Confined in Columns A through C, there are 3 clues that all use 1. Since each digit appears once in each column, each digit must appear exactly three times in any three columns. Thus, the 1 digit appears three times in Columns A through C, and we have already identified in which three clues they will be used. This means there cannot be any other 1s in Columns A through C.



This next part is hard to see. Consider the 6− clue in Column A. There are 3 options for a 6−: $(1, 7)$, $(2, 8)$, $(3, 9)$. Because we know there cannot be a 1, the $(1, 7)$ case is gone. Consider the $(3, 9)$ case. This means that the 12× in Column A is $(2, 6)$ because the 3 for $(3, 4)$ is taken. $A2$ is a 1 because the 3 in Column A is used in 6−. Recall that 48× in Row 1 is $(1, 6, 8)$ 2. In Column A, cell $A2$ is the 1 and the 6 is used in 12×. Then cell $A1$ is 8 and cells $A2$ and $A3$ are some permutation of $(1, 6)$.

Because cell $A2$ is 1, cell $B2$ is 3. Now consider the 8+ in Column B. There are also three choices for this: $(1, 7)$, $(2, 6)$, $(3, 5)$. Because we have used up all the 1s, $(1, 7)$ is eliminated. We also used the 3 in the 3÷ so the 8+ must be $(2, 6)$. Then cell $A2$ is 1 because 6 is used in the 8+.

This is the last part of this hypothetical. Recall that 70× is $(1, 2, 5, 7)_4$. Looking back at Column A, the 1 is in cell $A2$ and the 2 is in the 12× so cells $A6$ and $A7$ are 5 and 7 in some order. Thus 1 and 2 are in cells $B7$ and $C7$. However, in Column B, the 1 is in cell $B1$ and the 2 is used in 8+. Thus, there is no possible digit for cell $B7$, and this entire hypothetical is impossible.

Thus, the 6− that started all this in Column A cannot be $(3, 9)$ so it must be $(2, 8)$.

This is the fourth major realization of the puzzle.

Part V

We will now use what we just learned. Because the 6− is $(2, 8)$, the 12× cannot have a 2, so it must be $(3, 4)$. This means the 3 in Column A is used in 12×. Thus, cell $A2$ is a 1, and cell $B2$ is a 3.

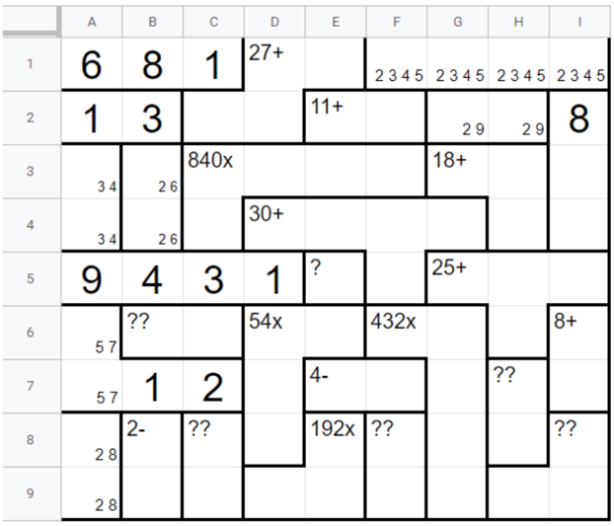
Inspect Column A. We have used the 1 in cell $A2$, the 2 and 8 in cells $A8$ and $A9$, and the 3 and 4 in cells $A3$ and $A4$. The remaining cells $(A1, A5, A6, A7)$ are filled with the four remaining digits $(5, 6, 7, 9)$. Notice that 9 cannot be in 12× or 70×. Thus, 9 must be in the 17+, so cell $A5$ is a 9. Now that cell $A5$ is taken, cell $A1$ must be the 6.

Recall the 8+ in Column B_4. We know there is no 1, and the 3 is used in cell $B2$ so 8+ must be $(2, 6)$ because $(1, 7)$ and $(3, 5)$ are impossible. Thus, cell $B7$ is a 1, and cell $C7$ is a 2. Then in $48\times$ in Row 1, cell $B1$ is a 8, and cell $C1$ is a 1.

Notice that in Column B we have found the precise locations of the $1, 3, 8$. We also know that the 8+ is made up by $(2, 6)$. The remaining four cells $(B5, B6, B8, B9)$ are filled by the remaining four digits $(4, 5, 7, 9)$.

Consider the 17+ in Row 5. Because cell $A5$ is a 9, the remaining three cells $(B5, C5, D5)$ add up to 8. Simply considering the remaining possibilities for Column B restricts cell $B5$ to be $4, 5, 7$, or 9. It of course cannot be 9. It also cannot be 7 because the two remaining cells $C5$ and $D5$ would have a sum of 1. Furthermore, it cannot be 5 because then cells $C5$ and $D5$ have a sum of 3. Although it is possible to achieve a sum of 3 in two cells, it would have to be made up of 1 and 2, both of which are used in Column C already. Thus, cell $B5$ is a 4.

Then $C5 + D5 = 4$, which has one solution, $(1, 3)$. The 1 in Column C is in $C1$, so cell $C5$ is a 3, and cell $D5$ is the 1.



This is the fifth major realization of the puzzle.

Part VI

Consider the $54\times$ clue that spans Columns D and E. Notice that because of its L shape, it allows you to repeat a digit twice in cell $E6$. What are the possibilities for a $54\times$ in four cells in this shape?: $(1, 1, 6, 9), (1, 2, 3, 9)$, and $(1, 3, 3, 6)$. Notice all of these options force a 1. Then there must be a 1 in the $54\times$. There is already a 1 in cell $D5$, so the 1 must go in cell $E6$.

Out of the three quadruples, only one can exist. $(1, 1, 6, 9)$ is eliminated because there cannot be a second

1 in Column D. $(1, 3, 3, 6)$ is also eliminated since 1 occupies cell $E6$, forcing both 3s into Column D. Thus, $54\times = (1, 2, 3, 9)$.

This is the sixth major realization of the puzzle.

Part VII

Return to Row 1. The digits in $48\times$ are 1,6,8. The digits in 14+ are 2,3,4,5. Thus, cells $D1$ and $E1$ are 7 and 9. We know there is a 9 in Column D in the $54\times$ clue. Thus, cell $D1$ is 7 and cell $E1$ is 9.

Consider Row 3. Where can the 9 go? Cell $A3$ is a 3 or 4; cell $B3$ is 2 or 6; 9 is not a factor of 840 so it cannot be cells $C3, D3, E3, F3$; with 8 in cell $I2$, $I3 + I4$ add to 9, so cell $I3$ cannot be 9. Therefore, 9 is in cell $G3$ or cell $H3$.

We have deduced the $18\times$ in Row 2 is $(2, 9)_1$. Thus, there are already two 9s in Columns G and H, so there cannot be any other 9s in these two columns.

Consider Row 4. Where can the 9 go? We just deduced that 9 cannot be in Columns G and H. Using similar logic as for Row 3, the 9 cannot be in cells $A4, B4, C4$ or $I4$. Then the only choices left for the 9 in Row 4 are in cells $D4, E4$, and $F4$.

Columns D and E have 9s already. Remember that the position of the 9 in column D is not precisely known, but we know it is in the $54\times$ clue 6. Since these two columns have 9s, the 9 in row 4 must be in cell $F4$.

You can use this same logic to find a 9 in one of the remaining columns where the position of the 9 is not precisely known (B, C, D, G, H, I). Try to find another 9. The answer is at the end of the article[1].

This is the seventh major realization of the puzzle.

Part VIII

Consider the $840\times$ that spans Rows 3 and 4. The prime factorization of 840 is $840 = 2^3 \cdot 3 \cdot 5 \cdot 7$.Notice that two of these five cells must be occupied by 5 and 7. The remaining three cells are occupied by digits that multiply to 24. Notice that because the $840\times$ spans two rows, there can be two of a number. However, this number cannot be 1, 2, or 3 because these three digits already occupy positions in Column C, where the repeated digit would appear.

A $24\times$ spans three cells and allows one repeat which is greater than 3, so it can be filled with (1,3,8), (1,4,6), or (2,3,4). Call the sum of the digits of whichever triplet is the correct triplet X. Let Y be the value of cell $F5$.

In Rows 3 and 4, the sum of these digits is $2 \cdot 45 = 90$. In Column A we have a sum of 7 across Rows 3 and 4. In Column B we have a sum of 8. There is guaranteed to be a 5 and a 7 in the $840\times$. Thus, we can write $7 + 8 + (12 + X) + (30 - Y) + 18 + (17 - 8) = 90$ which simplifies to $X - Y = 6$.

Consider the possible values for X. If the triplet is (1,3,8), $X = 12$. If it is (1,4,6), $X = 11$. If it is (2,3,4), $X = 9$. The respective values for Y that solve $X - Y = 6$ are $Y = 6$, $Y = 5$, and $Y = 3$.

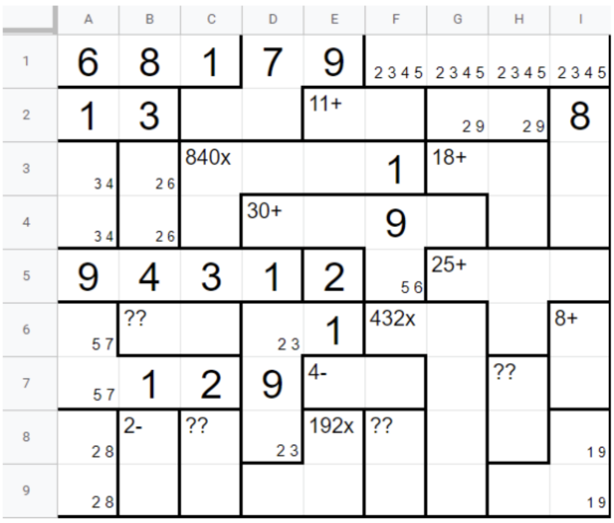
Y cannot be 3 because 3 is already in Row 5 in cell $C5$. The other two possibilities are still valid, however. This means that the corresponding triple (2,3,4) cannot be the one for 840×. Thus, the 840x must be (1,3,5,7,8) or (1,4,5,6,7). Either way, there is a 1. Notice Columns C, D, E all have 1s placed. Thus, the 1 is in cell $F3$.

Now consider the entire Column I. Where can the 1 go? The remaining three rows without 1s are Rows 4,8, and 9. However, the 1 cannot be in Row 4 because that would force 8 into cell $I3$ to complete 17+, but cell $I2$ is already an 8. Therefore, cell $I8$ or cell $I9$ is 1. The 9 in Column I cannot be in cell $I1$ because there is a 9 in cell $E1$, cell $I2$ is taken up by an 8, cells $I3$ and $I4$ add to 9 so neither can be the 9, cell $I5$ cannot be a 9 because 9 is in cell $A5$, 9 cannot be in cell $I6$ or cell $I7$ because the two cells add to 8. Thus, 9 is also in cell $I8$ or cell $I9$.

Because the 1 and 9 take up $I8$ and $I9$ in column I, the remaining 7 digits take up the other 7 spots. The digits in Column I add to 45. $I2 + I3 + I4 = 17$, $I6 + I7 = 8$, $I8$ and $I9$ are 1 and 9 in some order. Thus, $I1 + I5 = 45 - 17 - 8 - 1 - 9 = 10$.

Consider the remaining free cells in Row 5. We have narrowed down the possibilities for cell $F5$ to either 5 or 6. Notice that the 25+ cannot have any 9s because cell $A5$ is 9. Furthermore, we have deduced that columns G and H cannot have any more 9s. Thus, the max digit for any cell in the 25+ is 8. If the 2 in Row 5 is placed in the 25+, then the three remaining cells add to 23.

Because there are no 9s, the only way to add to 23 is (7,8,8). Because 8 is in $I2$, the 7 goes in cell $H5$, the 8s in cells $G5$ and $H6$, and the 2 in cell $I5$. Recall that $I1 + I5 = 10_{8}$. This makes cell $I1 = 8$, but 8 is in cell $I2$, so this is a contradiction. Thus, the 2 is not in the 25+, so it is in cell $E5$.



This is the eighth major realization of the puzzle.

Part IX

At this point, there are many cells that are down to two choices. We can pick one of these and see which of the choices will lead to a contradiction.

For simplicity's sake, the cell in question is cell $H6$.

We have already proved that cell $F5$ is a 5 or a 6. Then the remaining cells in Row 5 are (6,7,8) or (5,7,8), which will have sums of 21 and 20, respectively. This would make cell $H6$ either a 4 or a 5.

Which would be more useful to test first? In Column A, we also have a 5 that could potentially be in Row 6. Thus, we should let cell $H6$ be 5.

Immediately, this makes cell $A6$ be 7 and cell $A7$ be 5. Notice in Column B that the 5,7,9 go in cells $B6, B8$, and $B9$. Since cell $A6$ is 7 and we assumed cell $H6$ is 5, cell $B6$ must be 9. Furthermore, in Column I, the three choices for an 8+ clue that spans two cells (1,7),(2,6),(3,5) has been reduced to one. (1,7) and (3,5) are both impossible because 1s and 5s already occur in both Rows 6 and 7. Because cell $C7$ is a 2, cell $I6$ must be the 2, and cell $I7$ must be the 6. Consequently, cell $D6$ is a 3, and cell $D8$ is a 2.

Let's consider the 432x clue with $432 = 2^4 \cdot 3^3$. We have already proved that there are no more 9s in Columns G and H [7]. There is also a 9 in Column F in cell $F4$. Then there must be no 9s in the 432x. Then we need two 3s and a 6 or two 6s and a 3. Since cell $D6$ is a 3, cell $F6$ must be a 6 because the number in this cell is the only digit in this clue that can repeat.

From our original assumption that cell $H6$ is a 5, we know the sum of the digits in cells $G5, H5$, and $I5$ is 20. Then out of (5,6,7,8) that are still remaining in Row 5, the 6 is not used. Then the 6 must be in cell $F5$. However, we just found that cell $F6$ is a 6. There cannot be two 6s in the same column, so this is a contradiction, and our original assumption was incorrect.

Thus cell $H6$ must be a 4 and cell $F5$ must be a 5.

This is the ninth major realization of the puzzle.[2]

Part X

One feature that I love about KenKens is that although you may be concentrating on one clue or one part of the board, it will have influence over cells somewhere else.[3]

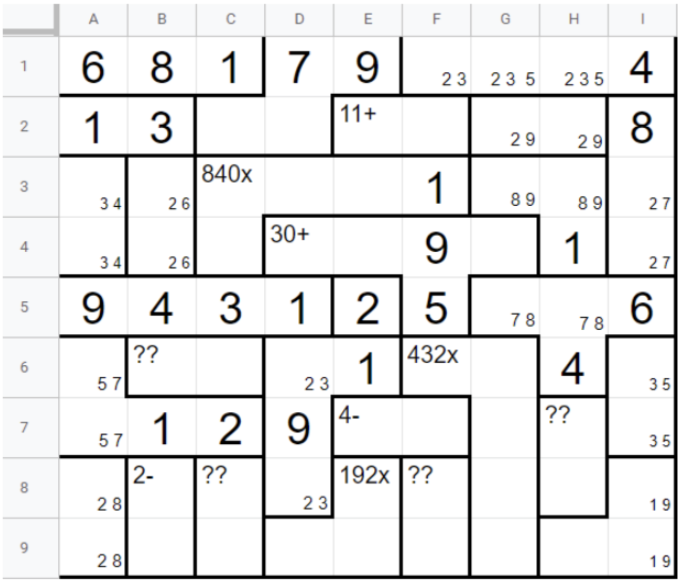
Where was cell $H6$ useful? It helped find cell $F5$. Where was cell $F5$ useful? If you recall, cell $F5$ was our value of Y [8]. Now we know that X is 11, so the digits in the 840x clue are (1,4,5,6,7).

Since the 840x clue is (1,4,5,6,7), then there is no 8. Notice in Row 3 that we have not found a location for the 8. We have just knocked out cells $C3, D3$, and $E3$ as possible habitats for the 8. The 8 also definitely cannot be in cells $A3$ or $B3$, and there is an 8 in cell $I2$. Then the 8 must be in cell $G3$ or cell $H3$, both of which are in a 18+ clue.

Recall that the 9 in Row 3 is also in cell $G3$ or cell $H3$ [7]. Then the sum of these two cells must be 8+9=17. Then we know for sure that cell $H4$ is a 1.

Notice that in Rows 3 and 4, because the 840× clue is (1,4,5,6,7), we are using both 4s and both 6s in these two rows. Consider cell $I3$. Recall that the digits in cells $I3$ and $I4$ add to 9 [8]. Since we used both 4s and 6s in these two rows, neither digit can appear in this clue, so the 17+ must be (2,7,8).

Finishing up Column I, we still have digits 3,4,5, and 6. Only 6 can go in cell $I5$ so we can fill that in. Also notice that the only way to satisfy 8+ is (3,5). Then cell $I1$ must be the 4.



Part XI

Return to Row 3. Consider the digit 3. The only possible spot for the 3 is in cell $A3$ because 840× is taken up by $(1, 4, 5, 6, 7)$ [10]. Cell $A4$ must be a 4.

Recall the principle that in two columns, each digit appears exactly twice and no more than *twice* [4,7]. Then in Columns G and H, the digit 8 appears no more than twice. We proved there must be an 8 in cell $G3$ or cell $H3$ [10]. We also know that cell $G5$ or cell $H5$ is an 8 because these two are the only two empty spots left in the row. Then there cannot be any more 8s. There also cannot be any more 9s.

Consider the 432× clue where $432 = 2^4 \cdot 3^3$.There is a 9 in cell $F4$, and we just proved there cannot be any more 9s in Column G. So we need to achieve 3^3 without 9s. Then there must be either two 3s and a 6 or two 6s and a 3. Either way, whichever digit is repeated must appear in cell $F6$. What are the possibilities? If the repeat is a 3, we have (3,3,6,1,8) and (3,3,6,2,4), but we can eliminate the first because we just proved there cannot be an 8. If the repeat is a 6, we have (3,6,6,1,4). Therefore, there is confirmed to be a 3,4, and 6 in Column G within the bounds of the 432× clue.

Now consider Row 4. We have already determined the positions of the 1,4,9. We also know that cell $F5$ is a 5. What digits are left for this row?: 2,3,5,6,7, and 8. Notice that within Rows 3 and 4, there will be a 2 in Column B and there will be a 2 in Column I. Thus, there cannot be a 2 in the 30+ clue. Similarly, we can eliminate the 6 and 7 because one is guaranteed in Columns B and I, respectively, and the other is guaranteed in the 840× clue. Thus, the remaining numbers for the 30+ clue are 3,5,8.

There are already two 8s in Columns G and H so cell $G4$ cannot be an 8. Furthermore, we just proved there must be a 3 in the 432× clue in Column G. Therefore, cell $G4$ is a 5. Similarly, because there is confirmed to be a 3 in the 54× clue in Column D, cell $D4$ is the 8, and cell $E4$ is the 3. We are almost done!

Now that cell $G4$ is confirmed to be 5, let's inspect where the 7 in Column G can go. It cannot be in cell $G1$ because 14+ is (2,3,4,5). The 7 cannot be in cell $G2$ because 18× is (2,9). It cannot be in cell $G3$ because 18+ is (1,8,9). Cell $G4$ is taken. The 7 does not divide 432, so it cannot be in cells $G6, G7, G8$, or $G9$. Therefore, cell $G5$ is a 7.

Consequently, cell $H5$ is a 8, cell $H3$ is a 9, cell $H2$ is a 2, cell $G2$ is a 9, and cell $G3$ is a 8.

Furthermore, because 5s appear in Columns F and G, cell $H1$ must be the 5 in the 14+ clue.

We also proved that within cells $G6, G7, G8$, and $G9$, there must be a 3,4, and 6. Then cell $G1$ cannot be 3, so it must be 2. Then cell $F1$ is a 3. But then cell $F6$ cannot be a 3, so it must be a 6. Aren't KenKens nice? Clues loop back to solve each other!

The next list can be verified easily (with the clues you've obtained from reading 7 pages of solution up to this point!)

Cell $G6$ is a 3, cell $I6$ is 5, cell $I7$ is 3, cell $A6$ is 7, cell $A7$ is 5, cell $D6$ is 2, cell $D8$ is 3, cell $B6$ is 9, and cell $C6$ is 8.

Since cell $G6$ is a 3, cell $F6$ cannot be a 3, so cell $F6$ must be a 6.

In Row 7, the 4- clue must be (4,8) because (1,2,3,5) are all taken in the row. Then in Row 7, the 7 must be in cell $H7$ because 7 does not divide 432. This means the 6 is in cell $G7$.

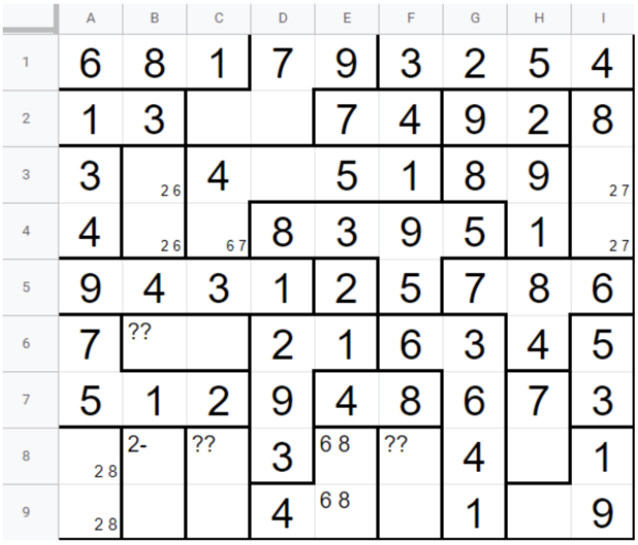
We will return where we started: Row 2. There, the 11+ clue is left to (4,7) or (5,6). Notice that in column F we have already used both 5 and 6 in cells $F5$ and $F6$ (coincidence?) respectively. This means that the 11+ clue must be (4,7).

Our favorite conclusion: in two columns, each digit appears exactly twice, so 4,7,11, will be reused in Columns E and F. We know there is a 4 in Row 2 and a 4 in Row 7. Then there is no 4 in Rows 8 or 9 in Columns E and F. How can we satisfy the 192× clue? $192 = 2^6 \cdot 3$. Because we already used both 3s in Columns D and E, there must be a 6. Then the other two numbers multiply to 25=32, so they must be 4 and 8. Then there must be a 4 in the 192× clue.

We just explained that there cannot be any more 4s in Columns E and F. Then cells $E8$ and $E9$ cannot be the 4. The 4 must be in cell $D9$. Because the 4 is in cell $D9$, cells $E8$ and $E9$ must be 6 and 8. This means that cell $E7$ cannot be 8, so cell $E7$ is 4.

More simple solving because you are a KenKen pro!!

Cell $F7$ is 8, cell $G8$ is 4, cell $G9$ is 1, cell $I8$ is 1, cell $I9$ is 9, cell $E2$ is 7, cell $F2$ is 4, cell $E3$ is 5, and cell $C3$ is 4.



Part XII

From here, we have used all known clues. Now is the time to bust out our pro Sudoku skills![4]

Column H is still missing a 3, but it can only go in cell $H8$ or cell $H9$. In Row 8, there is a 3 occupying cell $D8$. Therefore the 3 in Column H must go in cell $H9$. The last digit missing in Column H, a 6, goes in cell $H8$.

Likewise, cell $E9$ is a 6, and cell $E8$ is a 8. Cell $A9$ is a 9, and cell $A8$ is a 2. Cell $F9$ is a 2, and cell $F8$ is a 7. Cell $B9$ is a 7, and cell $B8$ is a 5. Just take a moment to admire this chain of events.

Row 9 is missing a 5 so it must go in the last empty cell, cell $C9$. Similarly, cell $C8$ is a 9.

From here, I must depart. As my last words to you, I know you are destined to go on and complete more great puzzles. Remember your training! Go you!

Part XIII

The remainder of the puzzle is left as an exercise to the reader.

	A	B	C	D	E	F	G	H	I
1	6	8	1	7	9	3	2	5	4
2	1	3			7	4	9	2	8
3	3	26	4		5	1	8	9	27
4	4	26	67	8	3	9	5	1	27
5	9	4	3	1	2	5	7	8	6
6	7	??		2	1	6	3	4	5
7	5	1	2	9	4	8	6	7	3
8	2	5	9	3	8	7	4	6	1
9	8	7	5	4	6	2	1	3	9

Notes

1) Answer: The 9 in Column D is in cell $D7$. In Row 7, there cannot be a 9 in cells $A7$ or $I7$ because they make the clue impossible to achieve. The digits 1 and 2 occupy cells $B7$ and $C7$. There are 9s in Columns E, F, G, H that are confirmed to not be in Row 7. Thus the 9 is in cell $D7$.

2) Although it may seem tedious and stumbled upon by chance, sometimes you will be forced to bash one option until you reach a contradiction. What's important is that we have the mindset that is willing to bash. Bashing is not a terrible technique. Furthermore, we have already limited the options of the cell from 9 down to 2, which is already tremendous work.

3) If you do Kakuro, you may notice that if two clues do not intersect, there is basically no relationship between the two. This is not the case with KenKens.

4) Sudoku is a game similar to KenKen. Usually there is a 9×9 grid separated into nine 3×3 grids. In each row, column, and 3×3 grid, each digit from 1 to 9 appears once. There are no arithmetic clues.